SCiO Systems Practitioner Competency Framework Resources

Competency	Books	Papers / articles / videos / presentations / WIKI
Section 1. Systems Knowledge & Skills - methods or methodologies that may be used in any conjunction, as part of a systemic inquiry into a problematic situation, and are rooted in systems thinking laws and concepts.		
Confrontation Analysis (CONAN)	Confrontation Analysis – How to win operations other than war, 1999, Howard Rational Analysis for a Problematic World Revisited 2e: Problem Structuring Methods for Complexity, Uncertainty and Conflict, 2001, Rosenhead The Utility of Force, 2007, Rupert Smith The Confrontation Analysis Handbook, 2018, Curry & Young	https://en.wikipedia.org/wiki/Confrontation_analysis
Critical System Heuristics	Systems Approaches to Making Change: A Practical Guide, 2020, Reynolds & Holwell	A Brief Introduction to Critical Systems Heuristics, 2005, Ulrich https://en.wikipedia.org/wiki/Critical_systems_thinking
INFORMED Group Dynamics		Organisational decision-making: A personal construct perspective, University of Manchester, UK, 2017, Marguet.N
Interactive Management	A Handbook of Iinteractive Management, 1994, Warfield & Cárdenas	Interactive Management (IM), Lee The Life and Work of Dr. John Warfield
Interactive Planning	Re-Creating the Corporation: A Design of Organizations for the 21st Century, 1999, Ackoff	A Brief Guide to Interactive Planning and Idealized Design, 2001, Ackoff https://en.wikipedia.org/wiki/Interactive_planning
Mosaic Transformation, Bubble Strategy	Fractal Organization, 2007, Hoverstadt	Mosaic Transformation, Fractal The Bubble Strategy: A Case Study of Dynamic, Defensible Processes of Change in Salford, 2007, Kawalek
Multi-methodology (incl. SOSM)	Rational Analysis for a Problematic World Revisited 2e: Problem Structuring Methods for Complexity, Uncertainty and Conflict, 2001 Rosenhead Systems Thinking – Creative Holism for Managers, 2003, Jackson Critical Systems Thinking and the management of Complexity, 2019, Jackson	
Patterns of Strategy	Patterns of Strategy, 2017, Hoverstadt & Loh	https://youtu.be/0Mi5F8z02Nk https://www.youtube.com/watch?v=irs7GiQyWLE&feature=youtu.be Patterns of Strategy
Socio-Technical Systems		The Evolution of Socio-Technical Systems, 1981, Trist The Social Engagement of Social Science, A Tavistock Anthology, Trist https://en.wikipedia.org/wiki/Sociotechnical_system

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Competency	Books	Papers / articles / videos / presentations / WIKI
Soft Systems Methodology	Systems Thinking, Systems Practice, 1981, Checkland Systems: Concepts, Methodologies, and Applications, 1990, Wilson Soft Systems Methodology: a 30-year retrospection, 1999, Checkland & Scholes Soft Systems Methodology: Conceptual Model Building and Its Contribution, 2001, Wilson Learning for Action: A Short Definitive Account of Soft Systems Methodology, and Its Use Practitioners, Teachers and Students, 2007, Checkland & Poulter Systems Approaches to Managing Change, 2010, Checkland Soft Systems Thinking, Methodology and the Management of Change, 2015, Wilson & van Haperen Systems Approaches to Managing Change, 2019, Checkland	https://en.wikipedia.org/wiki/Soft_systems_methodology
Syntegration/Syntegrity TM	Organizational Fitness: Corporate Effectiveness Through Management Cybernetics, 1992, Espejo & Schwanninger Beyond Dispute: The Invention of Team Syntegrity, 1994, Beer	http://www.syntegration.com https://en.wikipedia.org/wiki/Management_cybernetics#Syntegration
System Dynamics	Industrial Dynamics, 1961, Forrester Urban Dynamics, 1969, Forrester (N/A) Limits to Growth , 1972, Meadows, Managerial Applications of System Dynamics, 1981, Roberts. Elements of the System Dynamics Method, 1980, Randers The Fifth Discipline, 1990, Senge Business Dynamics: Systems thinking and modeling for a complex world, 2000, Sterman Limits to Growth: The 30-Year Update, 2004, Meadows Strategic Modelling and Business Dynamics: A Feedback Systems Approach, 2007, Morecroft. Strategic Management Dynamics, 2007, Warren Thinking in Systems: a Primer, 2008, Meadows Strategy Dynamics Essentials, 2015, Warren	System Dynamics Society: What is SD? https://en.wikipedia.org/wiki/System_dynamics

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Viable System Model	Brain of the Firm, 1973, Beer Designing Freedom, 1974, Beer Heart of Enterprise, 1979, Beer Diagnosing the System for Organizations, 1985, Beer The Viable System Model: Interpretations and Applications of Stafford Beer's VSM, 1989 Organizational Fitness: Corporate Effectiveness Through Management Cybernetics, 1992, Espejo & Schwanninger Fractal Organization, 2007, Hoverstadt Systemisches Management, 2009, Malik Systems Approaches to Managing Change, 2010, Hoverstadt Organizational Systems: Managing Complexity with the Viable System Model, 2011, Espejo Complexity Approach To Sustainability, A: Theory And Application, 2011, Espinosa & Walker Corporate Policy and Governance: How Organizations Self-organize, 2012, Malik (N/A) Baukybernetik, 2013, Frahm Die Intelligente Organisation: Das Playbook für organisatorische Komplexität, 2018, Lambertz Systems Approaches to Making Change, 2019, Hoverstadt The Viability of Organizations Vol. 1: Decoding the "DNA" of Organizations, 2019, Lassl The Viability of Organizations Vol. 2: Diagnosing and Governing Organizations, 2019, Lassl The Viability of Organizations Vol. 3: Designing and Changing Organizations, 2019, Lassl	https://en.wikipedia.org/wiki/Viable_system_model The SCiO Organisational Maturity Model The variety engineering method: analyzing and designing information flows in organizations, 2011, Rosenkrantz & Holten Management control systems: A variety engineering perspective, O'Gradya, Morlidge & Rousea

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Competency	Books	Papers / articles / videos / presentations / WIKI
Section 2. Systems Thinking Knowledge & Skills - laws and concepts that are fundamental to systems thinking practice and form the basis for systems knowledge and skills.		
Systemic thinking & systems concepts	Systems Thinkers, 2020, Ramage & Shipp	Systems Approaches to Making Change: A Practical Guide, 2020, Reynolds & Holwell Systems Thinking: What, Why, When, Where and How?, Goodman
Systems Laws	The Grammar of Systems: from Order to Chaos & Back, 2022, Hoverstadt	The Grammar of Systems, 2022, Hoverstadt (presentation)
Transposing Theory (YoYo model)	The World, The Flesh and The Metal: Prerogatives of Systems, 1965, Beer, nature, edition 205 Decision and Control, 1966, Beer	The systemic yoyo model: an introduction, 2010, Lin

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Competency	Books	Papers / articles / videos / presentations / WIKI
Section 3. Intervention Knowledge & Skills - approaches to intervening in an organisational or institutional situation, which may be used in any conjunction with systems knowledge and skills methods and methodologies.		
Action learning	Action Learning Handbook, 2003, Brockbank	What is Action Learning, World Institute for Action Learning https://en.wikipedia.org/wiki/Action_learning
Action research	The Action Research Planner: Doing Critical Participatory Action Research, 2013, Kemmis, McTaggart et al.	https://en.wikipedia.org/wiki/Action_research
Agile (project mgmt.)	Agile Project Management: Creating Innovative Products, 2009, Highsmith	Agile Project Management – Agilism versus Traditional Approaches, 2016, Fernandez & Fernandez
Appreciative Inquiry	Appreciative Inquiry: A Positive Revolution in Change, 2012, Cooperrider & Whitney	What is Appreciative Inquiry?, Cooperrider https://en.wikipedia.org/wiki/Appreciative_inquiry
Coaching	Coaching: Approaches and Perspectives, 2008, Knight	https://en.wikipedia.org/wiki/Coaching
Constellations		https://en.wikipedia.org/wiki/Constellation_model
Conversation mapping	Organisational learning II: Theory, Method and Practice, 1995, Argyris & Schon Conversations For Action and Collected Essays: Instilling a Culture of Commitment in Working Relationships, 2013, Flores & Letelier Flores conversation model in: Understanding Computers and Cognition: A New Foundation for Design, 1986, Winnograd & Flores	Conversations for Action and Collected Essays, Flores
Covert operations		Organisational change and covert processes, 2018, Saxena
Critical Social Learning Systems	Communities of Practice and Social Learning Systems, 2000, Wenger Social Learning Systems and Communities of Practice, 2010, Blackmore	Social Learning Systems for Collective Action: http://oro.open.ac.uk/37508/ https://helen.wilding.name/2011/02/20/key-concepts-related-to-critical-social-learning-system/
Culture mapping	Exploring Corporate Strategy, 2002, Johnson & Scholes	Management Models: Johnson and Scholes' Cultural Web, CMI
Data analysis incl. SPC, statistics	Normality and the Process Behaviour Chart, 2000, Wheeler Bayesian Data Analysis, 2013, Gelman, Carlin et al.	https://en.wikipedia.org/wiki/Statistics https://en.wikipedia.org/wiki/Statistical_process_control#Application_of_SPC

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Competency	Books	Papers / articles / videos / presentations / WIKI
	Statistical Process Control, 2018, Oakland & Oakland	
Decision structure design		https://en.wikipedia.org/wiki/Design_rationale
Deming	Out of the Crisis, 1986, Deming The New Economics for Industry, Government, Education, 2018, Deming & Cahill The Deming Management Method, 1988, Walton	Model of Sustainable Organisation (MoSO), 2010, MoSO Collective https://en.wikipedia.org/wiki/W._Edwards_Deming
Detecting & Managing undiscussables		Making the Undiscussable and Its Undiscussability Discussable, 1980, Argyris
Facilitation		What we Mean by Facilitation, Institute of Cultural Affairs https://en.wikipedia.org/wiki/Facilitation_(business)
Flawless consulting	Flawless Consulting: A Guide to Getting your Expertise Used, 2011, Block	
Fractal Enterprise Model & Capabilities (Bider)		Fractal Enterprise Model, Bider Fractal Enterprise Model and its Usage for Business Model Innovation, 2018, Bider & Perjons
Graphic facilitation, incl. visualisation	Visualising Business Transformation: Pictures, Diagrams and the Pursuit of Shared Meaning, 2020, Whitla	
Iceberg model	Thinking in Systems: a Primer, 2017, Meadows, ed. Wright	Iceberg Model, Ecochallenge.org
Influencing and mediation	Influence: The Psychology of Persuasion, 2007, Cialdini Confrontation Analysis – How to win operations other than war, 1999, Howard	Organisational decision-making: A personal construct perspective, 2017, Marguet.N, University of Manchester, UK.
Influence mapping		Influence, Stakeholder Mapping and Visualization, 2008, Walker, Bourne & Shelley
Interview technique		SOARA Interview Technique
Ladder of abstraction		Critical Thinking via the Abstraction Ladder, 1991, Seabury
Ladder of inference		https://en.wikipedia.org/wiki/Chris_Argyris

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Competency	Books	Papers / articles / videos / presentations / WIKI
		https://www.youtube.com/watch?v=V85ZbpiLvBw
Large group decision approaches	Challenging Strategic Planning Assumptions: theory, cases and techniques, 1981, Mason & Mitroff, New York, Wiley	Strategic Assumptions Surfacing and Testing, Institute for Manufacturing, University of Cambridge
Large group engagement processes		
Lean & 6 sigma	Lean Thinking: Banish Waste and Create Wealth in your Corporation, 2003, Womack & Jones Six Sigma: The Breakthrough Management Strategy Revolutionizing the World's Top Corporations, 2006, Harry & Schroeder	https://en.wikipedia.org/wiki/Lean_thinking https://en.wikipedia.org/wiki/Six_Sigma
Learning design & learning conversations	Learning Conversations: The Self-Organised Learning way to personal and organizational growth, 2013, Harri-Augstein E. S. and Thomas L. F.	https://en.wikipedia.org/wiki/Instructional_design
Linear argument technique	Visualising Business Transformation: Pictures, Diagrams and the Pursuit of Shared Meaning, 2020, Whitla	Making more meaningful models: A hands-on guide, 2017, Whitla
Listening & multiple perspectives		Systems-Centred®– Working with Differences Differently, 2020, SCiO
Managing deflection / resistance & challenge	Flawless Consulting: A Guide to Getting your Expertise Used, 2011, Block Discussing the Undiscussable: A Guide to Overcoming Defensive Routines in the Workplace, 2008, Noonan Overcoming Organizational Defenses: Facilitating Organizational Learning, 2018, Argyris	
Metaphors		Major Metaphors of Communication and some Constructivist Reflections on their Use, Cybernetics & Human Knowing, 2,1: 3-25, 1993, Krippendorff Metaphors of Organisation, 2001, Lawley The Persuasive Effects of Metaphor: A Meta-Analysis, *Human Communication Research*, 2006, Sopory & Dillard

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Neuro-Linguistic Programming (NLP)	Introducing NLP: Psychological Skills for Understanding and Influencing People, 2011, OConner	https://en.wikipedia.org/wiki/Neuro-linguistic_programming
Presentation design		https://en.wikipedia.org/wiki/Visual_communication
Productive conversations	Discussing the Undiscussable: A Guide to Overcoming Defensive Routines in the Workplace, 2008, Noonan Overcoming Organizational Defenses: Facilitating Organizational Learning, 2018, Argyris	Teaching Smart People How to Learn, 1991, Argyris The Left Hand Column, 2000, Ross & Kleiner https://www.fastcompany.com/659549/leading-ideas-surface-your-difficult-issues
Public speaking		https://en.wikipedia.org/wiki/Public_speaking
Questionnaire design		https://en.wikipedia.org/wiki/Questionnaire_construction
Scientific Theory & Evidence		https://en.wikipedia.org/wiki/Scientific_theory
Selling systems approaches	Flawless Consulting: A Guide to Getting your Expertise Used, 2011, Block	
Stakeholder analysis	Stakeholder Analysis A Complete Guide, 2019, Gerardus Blokdyk	https://en.wikipedia.org/wiki/Stakeholder_analysis
Strategic Options Development and Analysis (SODA)	Strategic Options Development and Analysis, Fran Ackermann and Colin Eden, pp. 139-200 of Systems Approaches to Making Change: A Practical Guide, 2020, ed. Reynolds & Holwell Strategic Options Development and Analysis, Fran Ackermann and Colin Eden, 2010, ResearchGate DOI: 10.1007/978-1-84882-809-4_4	Using Strategic Options Development and Analysis (SODA) to understand the simulation accessibility problem, 2022, Collins, Thavlphoke and Tako, Vol 74, Issue 10, Journal of the Operational Research Society A systemic investigation into the feasibility and desirability of Strategic Options Development Analysis as a societal impact evaluation tool in STEM research projects., 2023, Hodges, The Open University, Open Research Online. Cognitive Mapping and Strategic Options Development and Analysis (SODA), 2011, Georgiou, Wiley Online Library Surveying applications of Strategic Options Development and Analysis (SODA) from 1989 to 2018, 2020, Abuabara & Paucar-Caceres, Elsevier European Journal of Operational Research, at: ScienceDirect.com

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Competency	Books	Papers / articles / videos / presentations / WIKI
Training design		https://en.wikipedia.org/wiki/Instructional_design
Transactional Analysis (TA)	Games People Play: The Psychology of Human Relationships, 2010, Berne	https://en.wikipedia.org/wiki/Transactional_analysis
Trust mapping & metrics	Flores conversation model in: Understanding Computers and Cognition: A New Foundation for Design, 1986, Winnograd & Flores	Value by Design 2019 Hoverstadt & Loh [copy via patrick.hoverstadt@systemspractice.org] Patterns of Strategy [Products], Hoverstadt
Values mapping and integration	Value Networks and the True Nature of Collaboration, 2015, Allee.V et al.	Value Network Analysis and Value Conversion of Tangible and Intangible Assets, 2008, Allee.V Value by Design 2019 Hoverstadt & Loh [copy via patrick.hoverstadt@systemspractice.org]
Vanguard Method	Systems Thinking in the Public Sector: The Failure of the Reform Regime.... and a Manifesto for a Better Way, 2008, Seddon	Leading a Revolution in Management Practice, Vanguard
Verbal Behaviours (Rackham)	Behaviour Analysis in Training, 1977, Rackham	
Workshop design		How to Prepare a Workshop, 2021, Chandran Planning a Workshop: Organizing and Running a Successful Event, Mindtools